GCI WORLD 2026 - FINAL ASSIGNMENT - EGYPT MARKET

Predicting & Preventing Telecom Churn

An evidence-based retention strategy for Company A - Egyptian market context

120M

Egypt mobile subscriptions (2nd Africa)

49.56%

Training-sample churn label (balanced dataset, not real-world churn)

~$1.67M

Gross annual revenue protected before costs

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Egyptian Telecom Market Analysis

$3.9B
Market size 2025
CAGR 3.85% to 2034

120M
Mobile subscriptions
~105% penetration

<2%/mo
Industry churn benchmark
not comparable to label

EGP 100
Average subscriber spending
Customers are highly price-sensitive

Saturated Market

4 operators compete for 120M subscriptions at 105% penetration. Growth is share-shifting, so retention beats broad acquisition.

5G Opportunity

Egypt awarded 5G spectrum in late 2024. The active upgrade cycle lets a hardware strategy reduce churn and support 5G migration.

Company A Signal

49.56% is the proportion of customers who churned in the 31-60 day observation window of this balanced dataset. It is not Company A real-world churn. Dataset ARPU (~$50/month) likely reflects a premium or postpaid subset.

Sources: MEA Tech Watch 2025; IMARC 2025; Mordor Intelligence 2026; Reichheld & Schefter HBR 2000

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Dataset Overview & EDA

100,000
Customer records
Client + Record merged

100
Feature columns
on Customer_ID

49.56%
Dataset-window churn
31-60 days ahead

13
Model features
EDA + business logic

Churn Distribution

Retained: 50.44% (50,440 customers)
Churned: 49.56% (49,560 customers)

The label reflects balanced sample composition for model training, not actual base churn.

Top Numeric Associations with Churn (absolute Pearson r)

eqpdays (device age)	**11.27%**
hnd_price (handset value)	**10.32%**
totmrc_Mean (monthly subscription charge)	**6.86%**
mou_Mean (monthly usage (minutes))	**5.7%**
complete_Mean	**5.17%**
avg3mou	**4.95%**

eqpdays has the strongest individual linear association with churn (|r| = 0.113), but all correlations are modest, reinforcing the need for a multivariate model.

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Hypothesis Generation & Verification

All differences statistically significant - Chi-Square test, $p < 0.001$ | Egyptian market rationale included

VERIFIED

H1 Older Devices -> Higher Churn

Feature: device age (eqpdays)

Very Old	New
56%	41%

Egypt 5G context: aging 4G-only devices are structurally disadvantaged vs competitors upgrade offers.

VERIFIED

H2 Declining Usage -> Higher Churn

Feature: usage change (change_mou)

Declining	Stable
51%	47%

Egypt multi-SIM behavior: customers shift usage to a rival SIM before formally cancelling.

VERIFIED, NOT RECOMMENDED

H3 Lower Charges -> Higher Churn

Feature: monthly charge (totmrc_Mean)

Low Q1	High Q4
55%	44%

Discounting already-low-ARPU customers risks margin erosion without guaranteed prevention.

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Machine Learning Model - Selection, Training & Results

Model Selection Rationale

Binary churn classification. ROC-AUC is used because campaigns need to rank customers by risk, not only classify them. XGBoost is strong on tabular billing data and provides feature importance.

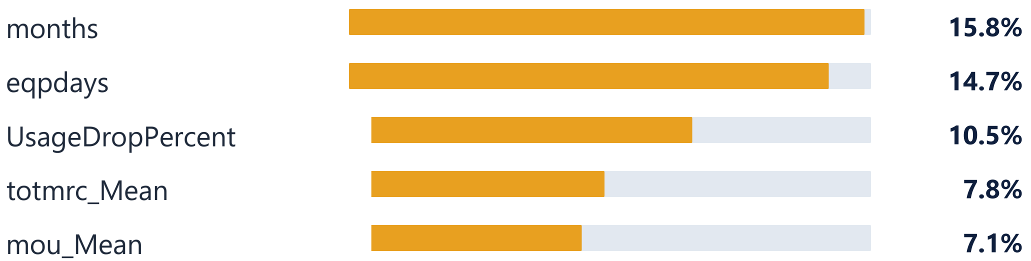
Engineered Features

UsageDropPercent: change_mou / (mou_Mean + 1)
RevenuePerMinute: rev_Mean / (mou_Mean + 1)

Model Comparison Results

Model	Accuracy	F1	ROC-AUC
Logistic Regression	57.5%	55.9%	59.9%
Random Forest	61.8%	62.6%	67.0%
XGBoost - Selected	**62.5%**	**62.3%**	**67.9%**

Top Churn Predictors (XGBoost)



ROC-AUC 0.679 means the model is useful for prioritizing outreach, not for confidently predicting a single customer. Campaign ROI requires hold-out validation.

Bottleneck Analysis

Most plausible churn pathway, consistent with ML results and Egyptian market dynamics.

Aging Hardware
eqpdays up - ML predictor

->

Poor User Experience
drop_vce up, custcare up

->

Declining Usage
change_mou < 0; multi-SIM switch

->

Customer Churn
near-term revenue loss

Egyptian Market Context

5G spectrum was awarded in late 2024, with commercial launches expected in H1 2025. Aging 4G devices are structurally disadvantaged, and competitor upgrade offers can accelerate churn.

Analytical Caution

Predictive importance is not causation. The hardware to disengagement mechanism should be validated with a hold-out control group and net ROI measurement.

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Candidate Proposals - Quantified Options

RECOMMENDED

A Device Upgrade Campaign

Target: device age > 342 days (median)
Segment: ~49,800 customers

Churn: 55.38%

Expected churners saved: 2,758

Gross revenue: $1.67M

Subsidized upgrade tied to 12-month contract; migrates customers to 5G network simultaneously.

B Usage Recovery Programme

Target: change_mou < 0 (declining)

Segment: ~53,700 customers

Churn: 51.20%

Expected churners saved: 2,747

Gross revenue: $1.92M

Personalized bundles to recapture usage before formal churn.

C Retention Discount

Target: totmrc_Mean Q1 (lowest)

Segment: ~28,800 customers

Churn: 54.97%

Expected churners saved: 1,584

Gross revenue: $0.64M

Risky in EGP inflation context; discounting low ARPU further erodes margins.

Assumption: prevent 10% of expected churners in each segment. Figures represent gross revenue protected before subsidy, incentive, and campaign costs; net ROI must be validated through pilot testing.

Why Proposal A Despite Lower Gross Revenue

Criterion	Proposal A	Proposal B	Winner
Gross Revenue	$1.67M	$1.92M	**B (+$0.25M)**
Top ML Predictor (eqpdays)	Directly targets	Indirect	**A**
Actionability	Hardware upgrade, clear offer	Usage recovery, behavioral	**A**
Strategic Value	5G migration + retention	Retention only	**A**
Recommended	A	--	**A**

Decision Rationale

B generates more gross revenue, but A is more defensible: it targets the #1 model signal, offers a concrete product lever, and bundles network modernization. Higher gross revenue without addressing hardware aging risks customer dissatisfaction.

Expected Business Impact - Proposal A

~$1.67M

Gross annual revenue protected before subsidy and campaign costs

Gross revenue by proposal

A is not chosen because it has the biggest gross number; it is chosen because it is the be targeted, more actionable pilot, pending net ROI validation.

Egypt-Specific Implementation Roadmap

1 Precision targeting

Score subscribers via XGBoost. Prioritize eqpdays > 342 plus high P(churn) and/or declining usage.

2 Deploy 5G upgrade offer

Subsidized 4G to 5G device, 30-50% off, contingent on 12-month contract renewal.

3 Prioritize by risk score

Contact highest predicted churn probability first to maximize ROI within budget constraints.

4 A/B test and measure

Hold-out control group. Track 90-day churn, uptake, subsidy cost, and net ROI.

All revenue estimates shown are gross revenue protected and require post-implementation ROI validation after subsidy and campaign costs.

Note: Revenue figures are for targeted high-ARPU segment (~$50/month ARPU).

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References & Citations

1. MEA Tech Watch (2025). Egypt Reaches 120 Million Mobile Phone Subscriptions in 2024.

2. IMARC Group (2025). Egypt Telecom Market Size, Share, Growth & Forecast 2034.

3. Mordor Intelligence (2026). Egypt Telecom MNO Market Size, Share & Industry Trends 2031.

4. Reichheld, F. & Schefter, P. (2000). E-Loyalty. Harvard Business Review, 78(4), 105-113.

5. Verbeke, W. et al. (2012). New insights into churn prediction. EJOR, 218(1), 211-229.

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7. XGBoost: https://xgboost.readthedocs.io/

8. Claude AI (Anthropic). Used for notebook review, requirement checking, analytical feedback, and generating draft versions that were subsequently modified and finalized. Conversation: [https://claude.ai/share/c124d322-7d74-4f18-b0fe-4ec2542ba829]

Conclusion

Company A Churn Prediction & Retention Strategy (Egypt)

Deadline: June 19, 2026 | GCI World 2026 April

Predictive Power

XGBoost (ROC-AUC: 0.679) ranks subscribers by churn risk using historical billing and usage data.

Root Signal

Device aging (eqpdays) is the strongest actionable signal, supported by testing and Egypt 5G context.

Actionable Strategy

Target aging-device customers prioritized by model score and usage decline, with a 5G upgrade offer.

Quantified Impact

10% expected-churner reduction protects ~$1.67M gross annual revenue before campaign costs.

Recommended action: launch a measured Proposal A pilot with a hold-out control group and net ROI tracking.